Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

9 solution use cases · 23 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

Contents

1. AWS
    - Solution use cases (9)
        - Provision AWS networking and compute
        - Deploy and operate a web application on AWS
        - Back up and restore AWS resources
        - Patch and manage EC2 instances
        - Troubleshoot AWS connectivity and resources
        - Audit and enforce AWS security posture
        - Use AWS as a dynamic inventory source
        - Manage CloudFormation stacks with Ansible
        - Migrate on-premises VMs to AWS
    - Product use cases
        - Amazon AI services (3)
        - Amazon Web Services (12)
        - Amazon Web Services (5)
        - Amazon Web Services (3)

AWS

Provision and operate Amazon Web Services resources.

Solution use cases

Use Case: Provision AWS networking and compute

Description	Build the base of an AWS environment, a VPC with subnets, routing, security groups and EC2 instances, with a playbook instead of console clicks. The same job template creates a development copy or a production copy, and can remove it again when it is no longer needed.
Collection	`amazon.aws` (Certified), `cloud.aws_ops` (Validated)
Key content	• `cloud.aws_ops.ec2_networking_resources` — A role to create a basic networking environment for an EC2 instance. • `cloud.aws_ops.manage_ec2_instance` — A role to create or delete an EC2 instance with optional networking resources. • `cloud.aws_ops.manage_transit_gateway` — Creation/Deletion of transit gateway with vpc/vpn attachment • `amazon.aws.ec2_vpc_net` — Configure AWS Virtual Private Clouds • `amazon.aws.ec2_vpc_subnet` — Manage subnets in AWS virtual private clouds • `amazon.aws.ec2_security_group` — Maintain an EC2 security group • `amazon.aws.ec2_instance` — Create & manage EC2 instances
Notes	• Use the Terraform route instead when the organization already standardizes on Terraform for provisioning, see Run Terraform from Automation Platform. • Existing CloudFormation stacks can stay in place and be driven by Ansible, see Manage CloudFormation stacks. • For a full application on top of this network, see Deploy and operate a web application on AWS.

Use Case: Deploy and operate a web application on AWS

Description	Deliver a complete web application, compute, load balancing and a database, on AWS as one automated workflow. Application teams get a repeatable way to create, update and move the stack, and operations teams get a known layout to support.
Collection	`cloud.aws_ops` (Validated), `amazon.aws` (Certified)
Key content	• `cloud.aws_ops.webapp.webapp` — Webapp • `cloud.aws_ops.webapp.webapp_ha_aurora` — Configure inventory for High availability Aurora cluster • `cloud.aws_ops.webapp.migrate_webapp` — Migrate webapp • `cloud.aws_ops.deploy_flask_app` — Deploy flask app in AWS. • `amazon.aws.elb_application_lb` — Manage an Application Load Balancer • `amazon.aws.rds_instance` — Manage RDS instances

Notes	• `cloud.aws_ops.create_rds_global_cluster` creates an Aurora global cluster across two regions for cross-region database resilience. • Pair with Back up and restore AWS resources so the database and volumes are protected from day one. • Treat the sample Flask role as a pattern; replace it with the organization's own deployment role.

Use Case: Back up and restore AWS resources

Description	Make backup policy a version-controlled part of the AWS environment. Plans, vaults and the resources they cover are created by automation, snapshots can be taken before risky changes, and restores are rehearsed so recovery is a known procedure.
Collection	`amazon.aws` (Certified), `cloud.aws_ops` (Validated)
Key content	• `cloud.aws_ops.backup_create_plan` — A role to create a backup plan and optionally a vault. • `cloud.aws_ops.backup_select_resources` — A role to configure backups for selected resources. • `amazon.aws.backup_vault` — Manage AWS Backup Vaults • `amazon.aws.backup_restore_job_info` — List information about backup restore jobs • `amazon.aws.ec2_snapshot` — Creates a snapshot from an existing volume • `amazon.aws.rds_instance_snapshot` — Manage Amazon RDS instance snapshots
Notes	• Related restore playbooks for audit trails and encryption keys are described in Audit and enforce AWS security posture. • Pair with Event-Driven Ansible to take an on-demand snapshot when a change request is approved. • Backup tags can be audited with `amazon.aws.backup_tag_info`.

Use Case: Patch and manage EC2 instances

Description	Keep EC2 instances patched and consistently configured. Existing servers are updated in place over AWS Systems Manager, so no inbound SSH is needed, and fleets built from AMIs are updated by baking a new image and replacing instances.
Collection	`amazon.aws` (Certified), `cloud.aws_ops` (Validated)
Key content	• `amazon.aws.aws_ssm` — connect to EC2 instances via AWS Systems Manager • `amazon.aws.aws_ec2` — EC2 inventory source • `amazon.aws.ec2_instance` — Create & manage EC2 instances • `amazon.aws.ec2_ami` — Create or destroy an image (AMI) in EC2 • `cloud.aws_ops.customized_ami` — A role to create custom AMI on AWS. User can create, update or delete AMI. • `amazon.aws.autoscaling_instance_refresh` — Start or cancel an EC2 Auto Scaling Group (ASG) instance refresh in AWS

Notes	• Operating-system patching tasks are covered in more depth under the Linux use cases, for example Red Hat Enterprise Linux with the standard package modules. • Use `amazon.aws.ec2_instance` with a stopped state before resizing or changing attached volumes. • Pair with Back up and restore AWS resources to take snapshots before patching.

Use Case: Troubleshoot AWS connectivity and resources

Description	When a workload in AWS cannot reach a destination, the cause is usually spread across route tables, security groups, network ACLs, gateways or peering. Run a playbook that walks the path between a source and a destination and reports which rule blocks the traffic.
Collection	`cloud.aws_troubleshooting` (Validated)
Key content	• `cloud.aws_troubleshooting.connectivity_troubleshooter` — connectivity_troubleshooter • `cloud.aws_troubleshooting.connectivity_troubleshooter_igw` — connectivity_troubleshooter_igw • `cloud.aws_troubleshooting.connectivity_troubleshooter_nat` — connectivity_troubleshooter_nat • `cloud.aws_troubleshooting.connectivity_troubleshooter_peering` — connectivity_troubleshooter_peering • `cloud.aws_troubleshooting.troubleshoot_rds_connectivity` — troubleshoot_rds_connectivity • `cloud.aws_troubleshooting.eval_security_groups` — Evaluate ingress and egress security group rules
Notes	• The `cloud.aws_ops` collection also ships `validate_route_tables`, `validate_network_acls` and `validate_security_group_rules` modules for the same kind of checks. • Pair with Event-Driven Ansible to run the diagnosis automatically when a monitoring alert reports a connection failure. • To fix what is found, see Provision AWS networking and compute.

Use Case: Audit and enforce AWS security posture

Description	Define the expected state of the account's security controls in code, such as the password policy, audit trails, encryption keys and security group rules, and use automation to report and correct differences. Controls that are disabled or deleted can be restored with a single job.
Collection	`amazon.aws` (Certified), `cloud.aws_ops` (Validated)

Key content	• `amazon.aws.cloudtrail` — manage CloudTrail create, delete, update • `amazon.aws.kms_key` — Perform various KMS key management tasks • `amazon.aws.iam_password_policy` — Update an IAM Password Policy • `amazon.aws.ec2_security_group` — Maintain an EC2 security group • `cloud.aws_ops.aws_restore_cloudtrail` — Include 'cloud.aws_ops.awsconfig_multiregion_cloudtrail' and 'cloud.aws_ops.enable_cloudtrail_encryption_with_kms' roles • `cloud.aws_ops.aws_restore_kms_key` — Cancel the deletion of the KMS key and re-enable the KMS key • `cloud.aws_ops.validate_security_group_rules` — Evaluates Security group Rules
Notes	• Pair with Event-Driven Ansible using the `amazon.aws.aws_cloudtrail` event source to react to events such as a trail being stopped or a security group being opened. • Keep enforcement behind an approval step for production accounts. • For secrets, `amazon.aws.secretsmanager_secret` and `amazon.aws.ssm_parameter` lookups keep credentials out of playbooks. • Related: Troubleshoot AWS connectivity and resources.

Use Case: Use AWS as a dynamic inventory source

Description	Let AWS itself be the source of truth for which servers exist. An inventory source syncs EC2 instances, and optionally RDS databases, into Automation Platform, so job templates always target current resources grouped by tag, region or account.
Collection	`amazon.aws` (Certified)
Key content	• `amazon.aws.aws_ec2` — EC2 inventory source • `amazon.aws.aws_rds` — RDS instance inventory source • `amazon.aws.sts_assume_role` — Assume a role using AWS Security Token Service and obtain temporary credentials • `amazon.aws.aws_ssm` — connect to EC2 instances via AWS Systems Manager
Notes	• Pair with Event-Driven Ansible and the `amazon.aws.aws_sqs_queue` or `amazon.aws.aws_cloudtrail` event source to run a configuration job when a new instance launches. • Use `compose` options to set `ansible_host` to the private or public address, depending on how the execution environment reaches AWS. • See Patch and manage EC2 instances for an example consumer of this inventory.

Use Case: Manage CloudFormation stacks with Ansible

Description	Many AWS estates already have CloudFormation templates (CFTs). Ansible can run those stacks as they are, so creation, updates and deletion go through the same job templates, approvals and credentials as the rest of the automation. Where templates become hard to maintain, the same resources can be moved to native `amazon.aws` modules over time.
Collection	`amazon.aws` (Certified)

Key content	• `amazon.aws.cloudformation` — Create or delete an AWS CloudFormation stack • `amazon.aws.cloudformation_info` — Obtain information about an AWS CloudFormation stack • `amazon.aws.s3_object` — Manage objects in S3 • `amazon.aws.ec2_vpc_net` — Configure AWS Virtual Private Clouds • `amazon.aws.ec2_instance` — Create & manage EC2 instances
Notes	• Replacing templates with modules gives readable tasks and Ansible variables, and avoids maintaining a second template language; keep CloudFormation where a template is shared with teams that still depend on it. • Be careful when deleting a stack, as it removes the resources it created. Use termination protection and a manual approval for production. • Related: Provision AWS networking and compute shows the native-module approach.

Use Case: Migrate on-premises VMs to AWS

Description	Move a virtual machine from an on-premises KVM hypervisor to AWS. The automation clones the VM, uploads the disk image to S3, imports it as an Amazon Machine Image and launches an EC2 instance from it.
Collection	`cloud.aws_ops` (Validated), `amazon.aws` (Certified)
Key content	• `cloud.aws_ops.move_vm_from_on_prem_to_aws.move_vm_from_on_prem_to_aws` — A playbook to migrate an existing on prem VM running on KVM hypervisor to AWS • `cloud.aws_ops.clone_on_prem_vm` — A role to clone an existing on prem VM using the KVM hypervisor. • `cloud.aws_ops.import_image_and_run_aws_instance` — A role that imports a local .raw image into an Amazon Machine Image (AMI) and run an AWS EC2 instance. • `cloud.aws_ops.upload_file_to_s3` — Upload a local file to S3 • `amazon.aws.ec2_import_image` — Manage AWS EC2 import image tasks
Notes	• The validated playbook targets KVM sources; other hypervisors need a different conversion step before the import. • The same pattern exists for Google Cloud, see Migrate on-premises VMs to Google Cloud. • Large images take time to upload; run the job on an execution node with good bandwidth to AWS.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Amazon AI services `amazon.ai`

Certified Red Hat · version 1.0.1 · updated 2026-05-08 · 25 modules → 3 use cases

Use Case: Run generative AI on Amazon Bedrock

Description	List foundation models, run inference and manage AgentCore runtimes and their endpoints.
Collection	`amazon.ai` (Certified, Red Hat)
Key content	**Foundation models** `bedrock_foundation_models_info` **Model inference** `bedrock_invoke_model` **AgentCore runtimes** `bedrock_agentcore_runtime`, `bedrock_agentcore_runtime_info` **AgentCore runtime endpoints** `bedrock_agentcore_runtime_endpoint`, `bedrock_agentcore_runtime_endpoint_info`

Use Case: Deploy machine learning with Amazon SageMaker

Description	Manage SageMaker models, endpoints and configurations, images, model package groups, pipelines and code repositories.
Collection	`amazon.ai` (Certified, Red Hat)
Key content	**Models** `sagemaker_model`, `sagemaker_model_info` **Endpoints** `sagemaker_endpoint`, `sagemaker_endpoint_info` **Endpoint configurations** `sagemaker_endpoint_config`, `sagemaker_endpoint_config_info` **Images** `sagemaker_image`, `sagemaker_image_info` **Image versions** `sagemaker_image_version`, `sagemaker_image_version_info` **Model package groups** `sagemaker_model_package_group`, `sagemaker_model_package_group_info` **Pipelines** `sagemaker_pipeline`, `sagemaker_pipeline_info` **Code repositories** `sagemaker_code_repository`, `sagemaker_code_repository_info`

Use Case: Detect operational issues with DevOps Guru

Description	Manage resource collections and read insights.
Collection	`amazon.ai` (Certified, Red Hat)
Key content	**Resource collections** `devopsguru_resource_collection`, `devopsguru_resource_collection_info` **Insights** `devopsguru_insight_info`

Amazon Web Services `amazon.aws`

Certified Red Hat · version 12.0.0 · updated 2026-10-13 · 141 modules → 12 use cases

Use Case: Manage EC2 instances and images

Description	Launch and manage EC2 instances, spot and dedicated hosts, AMIs, key pairs, launch templates, tags and instance metadata.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**EC2 instances** `ec2_instance`, `ec2_instance_info` **Instance types** `ec2_instance_type_info` **Spot instances** `ec2_spot_instance`, `ec2_spot_instance_info` **Dedicated hosts** `ec2_dedicated_host`, `ec2_dedicated_host_info` **Placement groups** `ec2_placement_group`, `ec2_placement_group_info` **Launch templates** `ec2_launch_template`, `ec2_launch_template_info` **AMIs** `ec2_ami`, `ec2_ami_info` **Image imports** `ec2_import_image`, `ec2_import_image_info` **Key pairs** `ec2_key`, `ec2_key_info` **Tags** `ec2_tag`, `ec2_tag_info` **Instance metadata** `ec2_metadata_facts`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Scale compute with Auto Scaling

Description	Manage Auto Scaling groups, their instances and instance refreshes.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**Auto Scaling groups** `autoscaling_group`, `autoscaling_group_info` **Auto Scaling instances** `autoscaling_instance`, `autoscaling_instance_info` **Instance refreshes** `autoscaling_instance_refresh`, `autoscaling_instance_refresh_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage block storage and S3

Description	Create and attach EBS volumes and snapshots, manage S3 buckets and objects.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**EBS volumes** `ec2_vol`, `ec2_vol_info` **EBS snapshots** `ec2_snapshot`, `ec2_snapshot_info` **S3 buckets** `s3_bucket`, `s3_bucket_info` **S3 objects** `s3_object`, `s3_object_info`

Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Build VPC networking

Description	Create VPCs, subnets, route tables, gateways, endpoints, peering, VPN, transit gateways, network ACLs, ENIs and elastic IPs.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**VPCs** `ec2_vpc_net`, `ec2_vpc_net_info` **Subnets** `ec2_vpc_subnet`, `ec2_vpc_subnet_info` **Route tables** `ec2_vpc_route_table`, `ec2_vpc_route_table_info` **Internet gateways** `ec2_vpc_igw`, `ec2_vpc_igw_info` **Egress-only internet gateways** `ec2_vpc_egress_igw`, `ec2_vpc_egress_igw_info` **NAT gateways** `ec2_vpc_nat_gateway`, `ec2_vpc_nat_gateway_info` **DHCP option sets** `ec2_vpc_dhcp_option`, `ec2_vpc_dhcp_option_info` **Network ACLs** `ec2_vpc_nacl`, `ec2_vpc_nacl_info` **VPC endpoints** `ec2_vpc_endpoint`, `ec2_vpc_endpoint_info` **Endpoint services** `ec2_vpc_endpoint_service_info` **VPC peering** `ec2_vpc_peering`, `ec2_vpc_peering_info` **VPN gateways** `ec2_vpc_vgw`, `ec2_vpc_vgw_info` **VPN connections** `ec2_vpc_vpn`, `ec2_vpc_vpn_info` **Transit gateways** `ec2_transit_gateway`, `ec2_transit_gateway_info` **Transit gateway attachments** `ec2_transit_gateway_vpc_attachment`, `ec2_transit_gateway_vpc_attachment_info` **Network interfaces** `ec2_eni`, `ec2_eni_info` **Elastic IPs** `ec2_eip`, `ec2_eip_info` **Security groups** `ec2_security_group`, `ec2_security_group_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Balance load and manage DNS

Description	Manage application and classic load balancers, Route 53 zones, records, health checks and signing keys.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**Application load balancers** `elb_application_lb`, `elb_application_lb_info` **Classic load balancers** `elb_classic_lb`, `elb_classic_lb_info` **Route 53 zones** `route53_zone` **Route 53 records** `route53`, `route53_info` **Route 53 health checks** `route53_health_check` **Route 53 key signing keys** `route53_key_signing_key`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage identity, access and encryption

Description	Manage IAM users, groups, roles, policies, access keys, instance profiles, password policy, KMS keys and temporary credentials.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**IAM users** `iam_user`, `iam_user_info` **IAM groups** `iam_group` **IAM roles** `iam_role`, `iam_role_info` **Inline policies** `iam_policy`, `iam_policy_info` **Managed policies** `iam_managed_policy` **Access keys** `iam_access_key`, `iam_access_key_info` **Instance profiles** `iam_instance_profile`, `iam_instance_profile_info` **MFA devices** `iam_mfa_device_info` **Password policy** `iam_password_policy` **KMS keys** `kms_key`, `kms_key_info` **Assumed role credentials** `sts_assume_role`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Run serverless functions

Description	Create Lambda functions, layers, aliases, event mappings and policies, and invoke functions.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**Lambda functions** `lambda`, `lambda_info` **Function aliases** `lambda_alias` **Event mappings** `lambda_event` **Layers** `lambda_layer`, `lambda_layer_info` **Policies** `lambda_policy` **Function invocation** `lambda_execute`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Run managed databases on RDS

Description	Manage RDS instances, Aurora clusters, parameter, option and subnet groups, and snapshots.
Collection	`amazon.aws` (Certified, Red Hat)

Key content	**RDS instances** `rds_instance`, `rds_instance_info` **RDS clusters** `rds_cluster`, `rds_cluster_info` **Global clusters** `rds_global_cluster_info` **Engine versions** `rds_engine_versions_info` **Instance parameter groups** `rds_instance_param_group`, `rds_instance_param_group_info` **Cluster parameter groups** `rds_cluster_param_group`, `rds_cluster_param_group_info` **Option groups** `rds_option_group`, `rds_option_group_info` **Subnet groups** `rds_subnet_group` **Instance snapshots** `rds_instance_snapshot` **Cluster snapshots** `rds_cluster_snapshot` **Snapshot queries** `rds_snapshot_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Back up resources

Description	Define backup plans, vaults, selections and tags, and review restore jobs.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**Backup plans** `backup_plan`, `backup_plan_info` **Backup vaults** `backup_vault`, `backup_vault_info` **Backup selections** `backup_selection`, `backup_selection_info` **Backup tags** `backup_tag`, `backup_tag_info` **Restore jobs** `backup_restore_job_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Provision stacks and monitor activity

Description	Deploy CloudFormation stacks, record CloudTrail, create CloudWatch alarms, rules and log groups.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**CloudFormation stacks** `cloudformation`, `cloudformation_info` **CloudTrail** `cloudtrail`, `cloudtrail_info` **Metric alarms** `cloudwatch_metric_alarm`, `cloudwatch_metric_alarm_info` **Event rules** `cloudwatchevent_rule` **Log groups** `cloudwatchlogs_log_group`, `cloudwatchlogs_log_group_info` **Log metric filters** `cloudwatchlogs_log_group_metric_filter`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Discover AWS accounts, regions and secrets

Description	Look up account, region and zone details, service IP ranges, Secrets Manager and SSM parameters.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**Caller identity** `aws_caller_info` **Availability zones** `aws_az_info` **Regions** `aws_region_info` **Account attributes** lookup `aws_account_attribute` **Collection constants** lookup `aws_collection_constants` **Service IP ranges** lookup `aws_service_ip_ranges` **Secrets Manager secrets** lookup `secretsmanager_secret` **SSM parameters** lookup `ssm_parameter`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Use AWS inventory, connections and events

Description	Build inventory from EC2 and RDS, connect through Systems Manager and react to CloudTrail and SQS events.
Collection	`amazon.aws` (Certified, Red Hat)
Key content	**EC2 inventory** inventory `aws_ec2` **RDS inventory** inventory `aws_rds` **Systems Manager connection** connection `aws_ssm` **CloudTrail events** eda event source `aws_cloudtrail` **SQS queue events** eda event source `aws_sqs_queue` **Resource actions summary** callback `aws_resource_actions`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Amazon Web Services `cloud.aws_ops`

Validated Red Hat · version 4.0.0 · updated 2025-09-04 · 3 modules → 5 use cases

Use Case: Provision EC2 instances and networking

Description	Create and delete EC2 instances with networking, custom AMIs and imported images, terminate instances by tag, and manage VPC peering and transit gateways.
Collection	`cloud.aws_ops` (Validated, Red Hat)

Key content	**EC2 instances** role `manage_ec2_instance` **EC2 networking resources** role `ec2_networking_resources` **Custom AMIs** role `customized_ami` **Image import and instance launch** role `import_image_and_run_aws_instance` **Terminate instances by tag** role `ec2_instance_terminate_by_tag` **VPC peering** role `manage_vpc_peering` **Transit gateways** role `manage_transit_gateway` **Internet gateway removal** role `awsconfig_detach_and_delete_internet_gateway`

Use Case: Validate network configuration

Description	Evaluate network ACLs, route tables and security group rules.
Collection	`cloud.aws_ops` (Validated, Red Hat)
Key content	**Network ACL evaluation** `validate_network_acls` **Route table evaluation** `validate_route_tables` **Security group rule evaluation** `validate_security_group_rules`

Use Case: Deploy applications and data services

Description	Deploy web apps, API Gateway with Lambda, Aurora global clusters, and move objects between S3 buckets.
Collection	`cloud.aws_ops` (Validated, Red Hat)
Key content	**Flask app** role `deploy_flask_app` **API Gateway with Lambda** role `awsconfig_apigateway_with_lambda_integration` **Aurora global clusters** role `create_rds_global_cluster` **S3 object moves** role `move_objects_between_buckets` **Web app** playbook `webapp.webapp` **HA Aurora inventory** playbook `webapp.webapp_ha_aurora` **Web app migration** playbook `webapp.migrate_webapp` **S3 file upload** playbook `upload_file_to_s3`

Use Case: Migrate on-premises VMs to AWS

Description	Clone on-premises KVM VMs and move them to AWS.
Collection	`cloud.aws_ops` (Validated, Red Hat)
Key content	**On-premises VM cloning** role `clone_on_prem_vm` **Move VM to AWS** playbook `move_vm_from_on_prem_to_aws.move_vm_from_on_prem_to_aws`

Use Case: Back up and secure AWS accounts

Description	Define credentials, create backup plans, select backup resources, and restore CloudTrail logging and KMS keys.

Collection	`cloud.aws_ops` (Validated, Red Hat)
Key content	**Credentials setup** role `aws_setup_credentials` **Backup plans** role `backup_create_plan` **Backup selection** role `backup_select_resources` **Multi-region CloudTrail** role `awsconfig_multiregion_cloudtrail` **CloudTrail KMS encryption** role `enable_cloudtrail_encryption_with_kms` **Restore CloudTrail** playbook `aws_restore_cloudtrail` **Restore CloudTrail encryption** playbook `aws_restore_cloudtrail_encryption` **Restore KMS key** playbook `aws_restore_kms_key`

Amazon Web Services `cloud.aws_troubleshooting`

Validated Red Hat · version 4.0.0 · updated 2025-09-02 · 9 modules → 3 use cases

Use Case: Troubleshoot network connectivity

Description	Evaluate security groups, network ACLs, NAT, internet gateway routes and VPC peering to find why traffic is blocked.
Collection	`cloud.aws_troubleshooting` (Validated, Red Hat)
Key content	**Connectivity troubleshooter** role `connectivity_troubleshooter` **Internet gateway checks** role `connectivity_troubleshooter_igw` **Local network checks** role `connectivity_troubleshooter_local` **NAT checks** role `connectivity_troubleshooter_nat` **VPC peering checks** role `connectivity_troubleshooter_peering` **Validation checks** role `connectivity_troubleshooter_validate` **Security group evaluation** `eval_security_groups` **Network ACL evaluation** `eval_network_acls` **NAT network ACL evaluation** `eval_nat_network_acls` **Source and internet gateway route evaluation** `eval_src_igw_route` **VPC peering evaluation** `eval_vpc_peering` **Next hop type** `get_connection_next_hop` **Network ACL validation** `validate_network_acls` **Route table validation** `validate_route_tables` **Security group rule validation** `validate_security_group_rules`

Use Case: Troubleshoot RDS connectivity

Description	Check connectivity to Amazon RDS instances.
Collection	`cloud.aws_troubleshooting` (Validated, Red Hat)
Key content	**RDS connectivity** role `troubleshoot_rds_connectivity`

Use Case: Set up AWS credentials

Description	Define credentials for AWS modules.
Collection	`cloud.aws_troubleshooting` (Validated, Red Hat)
Key content	**Credentials setup** role `aws_setup_credentials`